

The Ins and Outs of Optimal Transport

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Outline

- 1 Monge-Kantorovich Problems
 - Monge Problem
 - Kantorovich Problem
- 2 Wasserstein distances and spaces
 - Wasserstein distances
 - Properties of Wasserstein distance
 - Displacement interpolation
- 3 Entropic regularization
 - Sinkhorn's algorithm
- 4 Applications

Why use optimal transport?

- Creates a framework to minimize resource loss in trade networks, traffic (Prof. Gong), etc.
- Defines a natural, geometric way to measure differences between probability measures called the Wasserstein distance
- Computationally efficient implementations of optimal transport

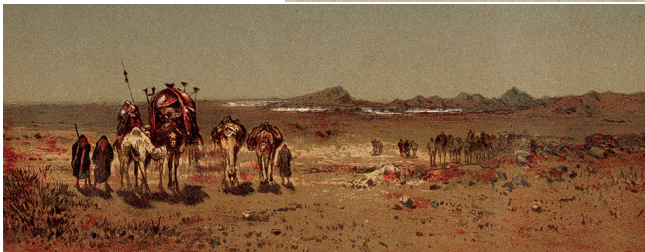
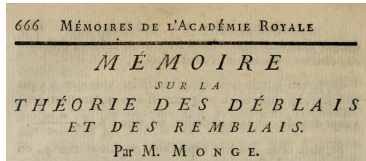
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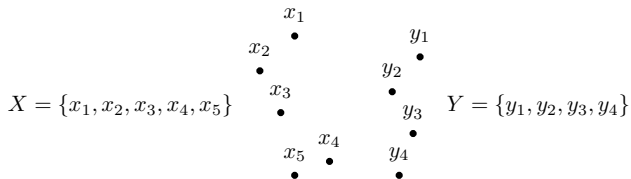
The origins of optimal transport



Gaspard Monge
1746-1818

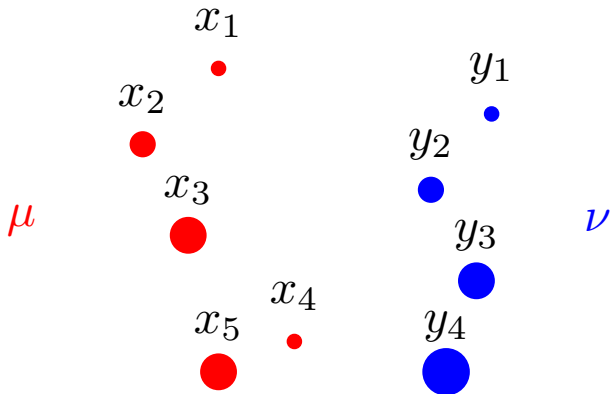


Example of the Monge problem



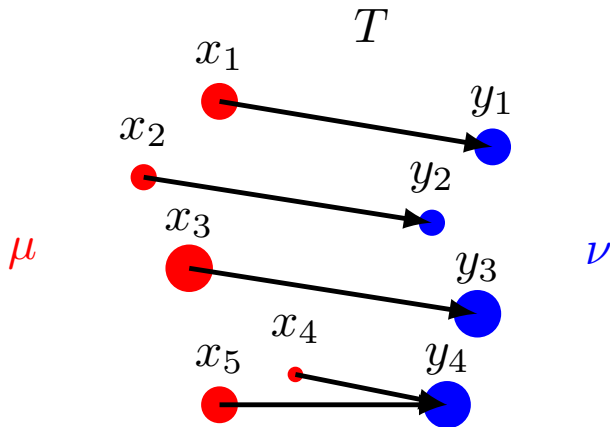
Start with sets of points X and Y .

Example of the Monge problem



Define the measure μ on set X and ν on set Y , which assign mass to every point.

Monge problem



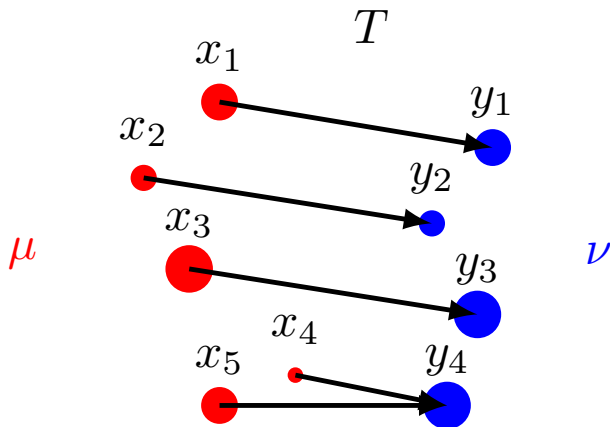
Construct a “mass-preserving” transport map that pushes the masses from X to Y .

What is optimal transport?

Optimal transport problems have three things in common:

- The starting and ending probability distributions.
- The cost function, which determines how much effort is spent moving mass from point A in the starting distribution to point B in the ending distribution. (Distance, money, human labor, energy or other metrics of effort)
- The optimal transport scheme, which specifies which masses from the starting distribution get moved where in the ending distribution.

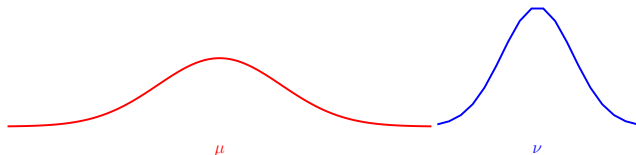
Example of the Monge problem



$$\begin{aligned} T_{\#}\mu(\{y_3, y_4\}) &= \mu(T^{-1}(\{y_3, y_4\})) = \mu(\{x_3, x_4, x_5\}) \\ &= \mu(x_3) + \mu(x_4) + \mu(x_5) = 0.7 = \nu(\{y_3, y_4\}). \end{aligned}$$

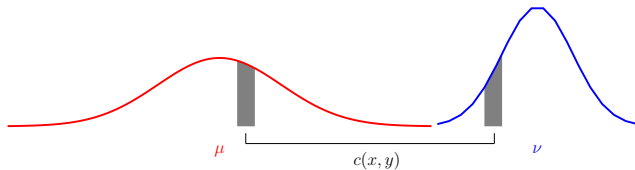
As expected, $T_{\#}\mu$ coincides with the measure ν .

Monge problem



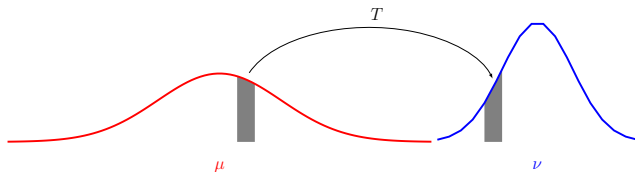
Consider two probability distributions (measures) μ and ν defined on metric spaces X and Y .

Monge problem



The cost function $c(x, y) : X \times Y \rightarrow [0, \infty]$ assigns a cost associated with travel between point $x \in X$ and $y \in Y$.

Monge problem



The transport map $T : X \rightarrow Y$ assigns points in X to Y such that *mass is conserved* when μ is moved to ν .

For $B \subseteq Y$,

$$T_{\#}\mu(B) = \mu(T^{-1}(B)).$$

To implement mass conservation, we require that

$$T_{\#}\mu = \nu.$$

Monge problem

Definition (Monge problem)

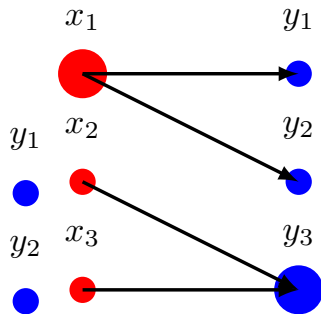
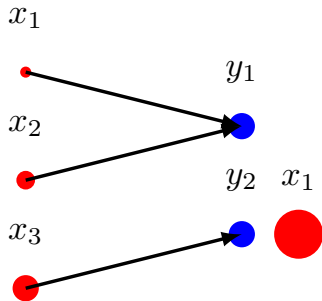
Given probability measures μ defined on a set X , ν defined on a set Y , and a cost function $c : X \times Y \rightarrow [0, \infty]$, find the transport map, a function $T : X \rightarrow Y$ that minimizes

$$\int_X c(x, T(x)) d\mu(x),$$

where the minimum is over functions $T : X \rightarrow Y$ satisfying $T_{\#}\mu = \nu$.

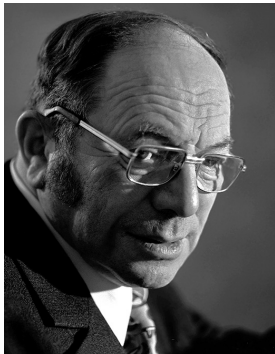
Intuitively, $\int_X c(x, T(x)) d\mu(x)$ is the total cost of moving one distribution to another according to a specific transport plan T .

Problems with the Monge problem



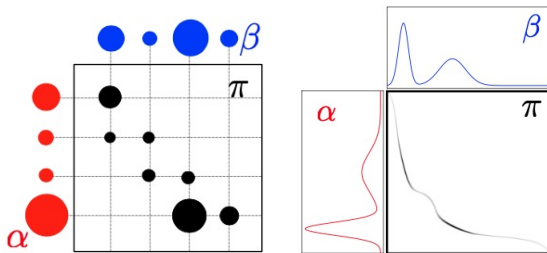
Splitting mass is not allowed in the Monge problem (because T is well defined), but is for the Kantorovich problem.

Kantorovich problem



Leonid Kantorovich reformulated optimal transport in the context of a resource transportation problem.

Kantorovich problem



π satisfies “conservation of mass”, i.e.

Kantorovich problem

Definition (Kantorovich problem)

Given probability measures μ defined on a set X , ν defined on a set Y , and a cost function $c : X \times Y \rightarrow [0, \infty]$, find the probability measure $\pi \in \Pi(\mu, \nu)$ for which the infimum in the following expression is attained:

$$\inf_{\pi \in \Pi(\mu, \nu)} \left\{ \int_{X \times Y} c(x, y) d\pi(x, y) \right\}.$$

The measure found in the Kantorovich problem is called an *optimal transport plan*.

Kantorovich problem

Question

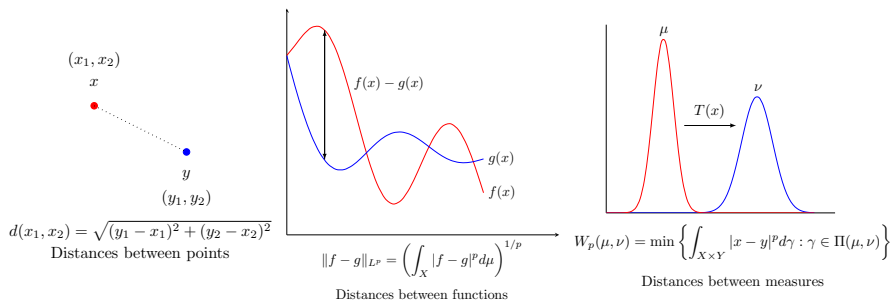
When do solutions to the Kantorovich problem exist?

Answer: Solutions exist for discrete optimal transport. For general continuous distributions, when the spaces X, Y are sufficiently “well-behaved” (Polish spaces) and the function is “smooth” (lower-semicontinuous). In the 1-dimensional setting, there is an explicit formula for optimal transport.

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Wasserstein distance



Comparing probability distributions is a fundamental problem in statistics and machine learning. It is not immediately obvious how to do this. Optimal transport provides a hint for defining a distance: before, we were interested in the optimal transport map or plan. What about the cost attained by the minimum optimal transport map or plan?

Wasserstein distance/space

Definition (Wasserstein space)

Consider the space of probability measures on a Polish metric space (X, d) with bounded p th moment for $p \geq 1$, i.e.

$$\mathcal{P}_p(X) := \left\{ \mu \in \mathcal{P}(X) : \int_X |x|^p d\mu(x) < \infty \right\}.$$

The p -**Wasserstein distance** is defined for $\mu, \nu \in \mathcal{P}_p(X)$ as the following:

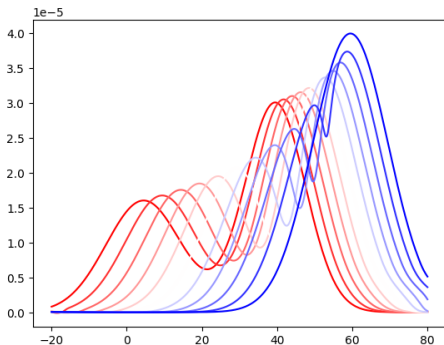
$$W_p(\mu, \nu) := \min_{\pi \in \Pi(\mu, \nu)} \left\{ \int_{X \times X} |x - y|^p d\pi \right\}^{1/p}.$$

In other words, $W_p(\mu, \nu)$ is the p th root of the minimum transport cost for the cost $c(x, y) = |x - y|^p$. The set and metric $(\mathcal{P}_p(X), W_p)$ constitute a **Wasserstein space**.

Properties of Wasserstein distance

- Wasserstein distance is a metric (distance function)
- Wasserstein spaces have interesting geometric structure

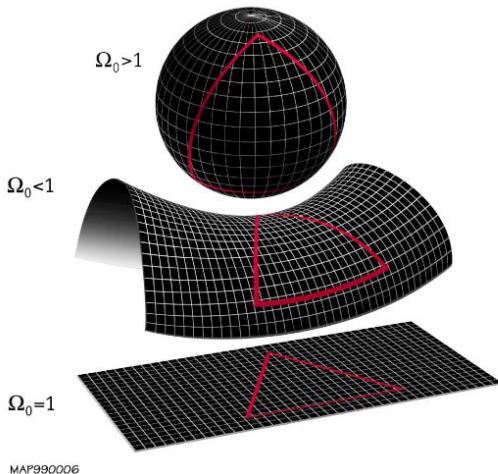
Adding time to optimal transport



What if we could use Wasserstein distances to continuously transform one probability distribution into another?

This process is called *displacement interpolation*.

Geodesics



Geodesics are the shortest paths connecting two points on metric spaces.

Wasserstein space is a geodesic space

Definition

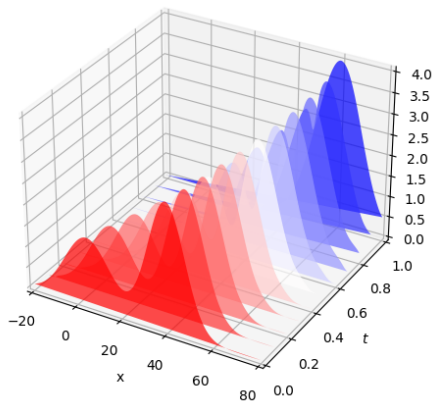
A geodesic space is a metric space if for every pair of points there exists a geodesic connecting them.

Theorem

The space $(\mathcal{P}_p(X), W_p)$ is a geodesic space. More specifically, define $\pi_t : X \times X \rightarrow X$ as $\pi_t(x, y) = (1 - t)x + ty$. Then the curve $\mu_t := (\pi_t)_\# \pi$ is a geodesic in $\mathcal{W}_p$ connecting $\mu_0 = \mu$ to $\mu_1 = \nu$ (where π is an optimal transport plan with marginals μ, ν).

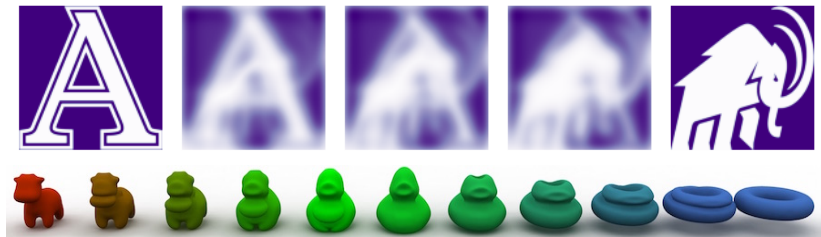
Interpolation along geodesics

Displacement interpolation



Modulating the parameter t for the geodesic curve μ_t allows one to travel along the geodesic between μ and ν .

Interpolation between images and surfaces



Why not try interpolating images and three-dimensional objects?

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Solving OT problems

- Network flow algorithms are generally slow in practice
- Curse of dimensionality: estimating continuous distributions with linear combinations of discrete distributions gets less accurate the higher the dimension

Entropic regularization

There are discrete equivalents to the terms in the Kantorovich problem, i.e., a cost matrix C_{ij} instead of a cost function $c(x, y)$, a matrix of weights P_{ij} instead of a joint probability measure μ , and a space $U(a, b)$ instead of the space of mass-preserving measures $\Pi(\mu, \nu)$.

Definition (Entropically regularized optimal transport)

Let $\varepsilon > 0$, $a \in \mathbb{R}^n$, $b \in \mathbb{R}^m$, C a cost matrix defined by a function $c(x, y)$ and $P \in U(a, b)$. Then the **entropically regularized optimal transport problem** is

$$L_C^\varepsilon(a, b) = \min_{P \in U(a, b)} \sum_{i,j} C_{ij} P_{ij} - \varepsilon H(P).$$

What's this new function $H(P)$?

Entropy

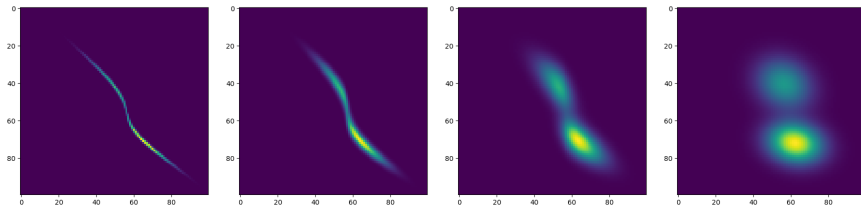
Definition (Entropy)

Let P be an admissible coupling. The **entropy** of P is defined as

$$H(P) = - \sum_{i,j} P_{ij} (\log(P_{ij}) - 1), \quad (1)$$

where $H(P) = -\infty$ if one of the entries P_{ij} is 0.

Adjusting the parameter ε



As ε increases, the transport plan for EROT becomes more smudged out.
 $H(P)$ is a *regularizer*, which prevent over/underfitting in machine learning.

For small ε , the coupling converges to the admissible coupling with
maximum entropy.

For large ε , the coupling converges to the maximum entropy distribution

$$\mu \otimes \nu.$$

Sinkhorn's algorithm

Theorem

The solution of the entropically regularized optimal transport problem is unique and has the form $P_{ij} = u_i K_{ij} v_j$ for all (i, j) , where u, v are positive real-valued vectors and K is a positive real-valued matrix.

We know K_{ij} a priori. To approximate u, v , we use **Sinkhorn's algorithm**.

Definition (Sinkhorn's algorithm)

$$u^{(\ell+1)} := \frac{a}{K v^{(\ell)}} \text{ and } v^{(\ell+1)} := \frac{b}{K^T u^{(\ell+1)}}.$$

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Some of the many applications of OT

- Geology (semigeostrophic equation)
- Economics (mechanism design)
- Control theory
- Machine learning (Wasserstein Generative Adversarial Networks)
- Computer graphics (Researched by Disney!)

OT is a hot field to be in!



Phillip Zhou



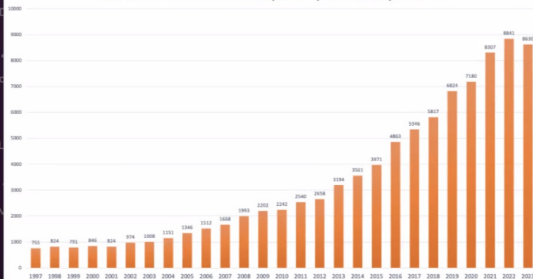
Jonah Botvinick-Greenh...



Yunan Yang

3 days ago in  group-discussion - OT_Trend.png

Research Activities on the Topic of Optimal Transportation



Thread



Yunan Yang 3 days ago

Hi all, I hope you had a restful break. Good luck with your plan to witness the solar eclipse tomorrow.

I had fruitful visits to MIT and Brown in the past few days. So many people are working on applications and extensions of optimal transport in various fields. Below is a plot I made mimicking a paper by Dejan Slepčev from 2016, in which the plot stopped halfway through 2016. It is impressive how large the community is now.

OT_Trend.png ▾



Thanks for listening!!!

Measure Theory Preliminaries

Measure theory studies measure spaces, i.e. a set, subsets of the set that can be assigned a measure/size, and a function that assigns each subset in the sigma-algebra a size). Formally,

Definition (Measure space)

A measure space is a triple $(\mathcal{X}, \Sigma, \mu)$ where $\mathcal{X}$ is a set, Σ is a σ -algebra, and μ is a measure.

Definition (Probability space)

A probability space is a measure space $(\Omega, \mathcal{F}, P)$, where Ω is the sample space, $\mathcal{F}$ is the event space, and P is a probability function.

Don't worry about these details at all, this is just to tell you that probability theory is founded on measure theory and the two are intimately related.

What is a measure?

You can think of a *measure* as a function that takes in a set and outputs the size, or volume, of the set. It satisfies “nice properties”, for instance, measure is positive, the empty set has zero measure, and measures of disjoint sets add. In mathematical terms, assuming (X, Σ, μ) is a measure space,

For all E in Σ , $\mu(E) \geq 0$.

$$\mu(\emptyset) = 0.$$

For pairwise disjoint sets $\{E_k\}$ in Σ , $\mu\left(\bigcup_n E_k\right) = \sum_n \mu(E_k)$.

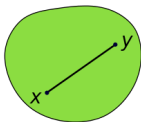
What is a convex set?

Assume we are operating in a Euclidean space, i.e. $\mathbb{R}^n$ with the standard inner product.

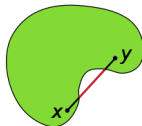
Definition

A **convex set** is a set C in a Euclidean space such that for any $x, y \in C$, the affine combination $(1 - \lambda)x + \lambda y$ belongs to C for all $\lambda \in [0, 1]$.

In other words, it is a set such that for any $x, y \in C$, the line segment containing x and y is also in C .



A convex set



A non-convex
set

Weak convergence

Definition (Weak convergence)

Let (X, Σ) be a measurable space and $(\mu_i)_{i=1}^\infty$ and μ be finite Borel measures on (X, Σ) . The sequence of measures $(\mu_i)_{i=1}^\infty$ **weakly converges** to μ (denoted $\mu_i \rightharpoonup \mu$) if

$$\int_X f d\mu_i \rightarrow \int_X f d\mu \text{ as } i \rightarrow \infty \text{ for all } f \in C_b(X),$$

where $C_b(X)$ is the space of continuous and bounded real-valued functions on X , i.e.,

$$C_b(X) := \{f : X \rightarrow \mathbb{R} : f \text{ is continuous and bounded}\}.$$

C_0 denotes the functions on X that vanish at infinity.

Monge problem

Definition (Monge problem)

Let (X, Σ_1) and (Y, Σ_2) be two measurable spaces. Given probability measures $\mu \in \mathcal{P}(X)$, $\nu \in \mathcal{P}(Y)$, and a function $c : X \times Y \rightarrow [0, \infty]$ called the “cost function”, find the transport map, a function $T : X \rightarrow Y$ that is the infimum of the quantity

$$M(T) := \int_X c(x, T(x)) d\mu(x),$$

where the infimum is taken over μ -measurable maps $T : X \rightarrow Y$ satisfying $T_{\#}\mu = \nu$.

Kantorovich problem

Definition (Kantorovich problem)

Let (X, Σ_1) and (Y, Σ_2) be two measurable spaces. Given probability measures $\mu \in \mathcal{P}(X)$, $\nu \in \mathcal{P}(Y)$, and a cost function $c : X \times Y \rightarrow [0, \infty]$ find the probability measure $\pi \in \Pi(\mu, \nu)$ for which the infimum in the following expression is attained:

$$\inf_{\pi \in \Pi(\mu, \nu)} \left\{ \int_{X \times Y} c(x, y) d\pi(x, y) \right\},$$

Pushforward measure

Definition (Pushforward measure)

Given measurable spaces (X, Σ_1) and (Y, Σ_2) where X and Y are separable metric spaces, $\mu \in \mathcal{P}(X)$, where $\mathcal{P}(\cdot)$ denotes the space of probability measures on a set, and a μ -measurable map $T : X \rightarrow Y$, the **pushforward measure** of μ by T denoted $T_{\#}\mu \in \mathcal{P}(Y)$ is defined by

$$(T_{\#}\mu)(B) = \mu(T^{-1}(B)) \text{ for all Borel } B \subseteq Y.$$

By definition, for any bounded Borel function (a measurable function defined on a Borel space) $f : Y \rightarrow \mathbb{R}$ we also have

$$\int_X f(T(x)) d\mu(x) = \int_Y f(y) d(T_{\#}\mu)(y).$$

This equality is known as the “change of variables” formula.

Convergence of EROT as $\varepsilon \rightarrow 0$

Proposition

The unique solution P_ε of Definition 6 converges to the optimal transport plan (within the set of all optimal solutions of the Kantorovich problem) with maximal entropy, i.e. as $\varepsilon \rightarrow 0$, $L_\varepsilon^\varepsilon(a, b) \rightarrow L_c(a, b)$, where L_c is defined in the discrete Kantorovich problem, and

$$P_\varepsilon \rightarrow \operatorname{argmin}_P \{-H(P) : P \in U(a, b), \langle P, C \rangle = L_c(a, b)\}. \quad (2)$$

As $\varepsilon \rightarrow \infty$, $P_\varepsilon \rightarrow a \otimes b = ab^T = (a_i b_j)_{i,j}$.

Proposition

The solution to Definition 6 is unique and has the form $P_{ij} = u_i K_{ij} v_j$ for all $(i, j) \in \{1, \dots, n\} \times \{1, \dots, m\}$, where $(u, v) \in \mathbb{R}_+^n \times \mathbb{R}_+^m$ and $K \in M_+^{n \times m}(\mathbb{R})$.

Characterization of coupling measures

Theorem (Characterization of coupling measures)

The following are equivalent:

- *For all measurable sets $A \subset X, B \subset Y$, we have $\pi[A \times Y] = \mu[A]$ and $\pi[X \times B] = \nu[B]$.*
- *For all integrable measurable functions φ, ψ on X and Y respectively,*

$$\int_{X \times Y} (\varphi(x) + \psi(y)) d\pi(x, y) = \int_X \varphi d\mu + \int_Y \psi d\nu.$$

1-D Kantorovich problem

Theorem (1-D Kantorovich Problem)

Let $\mu, \nu \in \mathcal{P}(\mathbb{R})$ with cumulative distribution functions F and G , respectively. Define a cost function $c(x, y) = d(x - y)$ where d is convex and continuous. Let $\pi^\dagger$ be the measure on $\mathbb{R}^2$ with cumulative distribution function $H(x, y) = \min\{F(x), G(y)\}$, or in other words, the co-monotone transport plan between μ and ν . Then $\pi^\dagger \in \Pi(\mu, \nu)$ and is optimal for the Kantorovich problem with the specified cost function c . Additionally, the optimal transportation cost is

$$\min_{\pi \in \Pi(\mu, \nu)} \int_{X \times Y} d(x - y) d\pi(x, y) = \int_0^1 d(F^{[-1]}(t) - G^{[-1]}(t)) dt.$$

CDFs and generalized inverses

Definition (Cumulative distribution function)

For a measure $\mu \in \mathcal{P}(\mathbb{R})$, its **cumulative distribution function**, or CDF, is defined as

$$F(x) := \int_{-\infty}^x d\mu = \mu((-\infty, x]), \forall x \in \mathbb{R}.$$

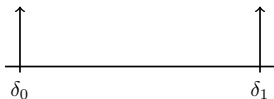
For a measure $\pi \in \mathcal{P}(\mathbb{R}^2)$, its **cumulative distribution function** is defined as

$$H(x_0, y_0) := \int_{R(x_0, y_0)} d\pi = \pi[R(x_0, y_0)],$$

where $R(x_0, y_0)$ is the rectangle consisting of all points $(x, y) \in \mathbb{R}^2$ such that $(x, y) \in (-\infty, x_0] \times (-\infty, y_0]$.

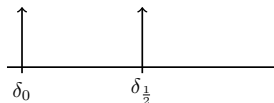
Properties of Wasserstein distance

- Wasserstein distance is a metric (distance function)
- Wasserstein distances metrize weak convergence
- Wasserstein spaces have interesting geometric structure



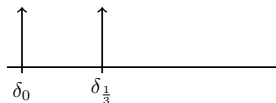
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The Kullback-Leibler divergence registers the distance between δ_0 and $\delta_{1/n}$ as infinity for all n , but as $n \rightarrow \infty$, the Wasserstein distance between δ_0 and $\delta_{1/n}$ weakly converges to zero.

Metrizing weak convergence means that weak convergence occurs if and only if convergence in Wasserstein distance occurs.

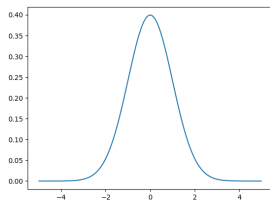
CDFs and generalized inverses

Definition (Generalized inverse of a CDF)

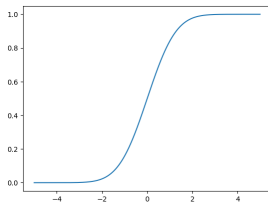
Let F be the cumulative distribution function of some measure μ . We define the **generalized inverse** of F as a function $F^{[-1]} : [0, 1] \rightarrow \mathbb{R}$,

$$F^{[-1]}(t) = \inf\{x \in \mathbb{R} : F(x) > t\}, \forall t \in [0, 1].$$

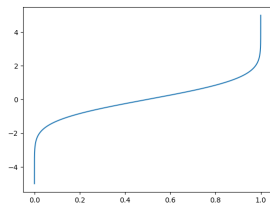
CDFs and generalized inverses



(a) Probability density function of the standard normal distribution $\mathcal{N}(0, 1)$.



(b) Cumulative distribution function of the standard normal distribution $\mathcal{N}(0, 1)$.



(c) The quantile function of the standard normal distribution $\mathcal{N}(0, 1)$.

Let $X \sim \mathcal{N}(0, 1)$. Plotted above is its probability density function, cumulative distribution function, and quantile function.

Dynamic optimal transport

Definition (Continuity equation)

$$\frac{\partial_t \rho_t}{\partial t} + \nabla \cdot (v_t \rho_t) = 0$$

$$A[\rho, v] = \int_0^1 E(t) dt = \int_0^1 \left(\int_{\mathbb{R}^n} \rho_t(x) |v_t(x)|^2 dx \right) dt$$

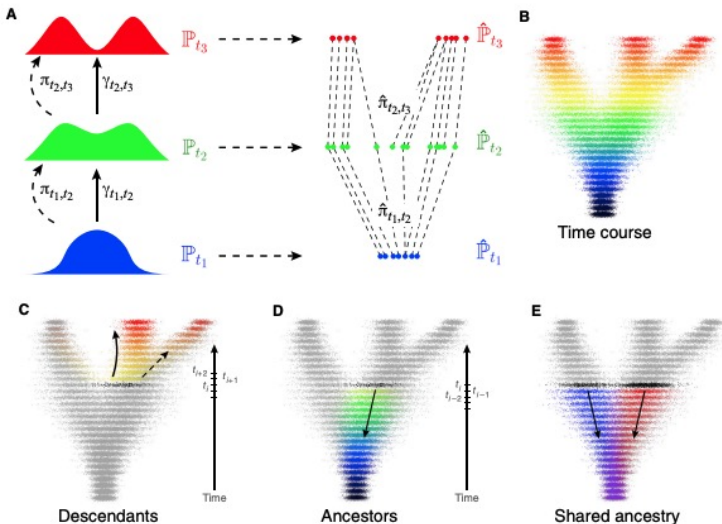
Theorem (Benamou-Brenier formula)

Given $\rho_0, \rho_1 \in \mathcal{P}(\mathbb{R}^n)$ having smooth densities with respect to L^n and bounded supports, then

$$W_2(\rho_0, \rho_1)^2 = \min_{(\rho_t, v_t)} A[\rho, v_t],$$

for (ρ_t, v_t) weakly satisfying (defined in Appendix) the continuity equation and where $\rho_0 = \mu, \rho_1 = \nu$.

Trajectory analysis



This uses both dynamic optimal transport and entropic regularization!